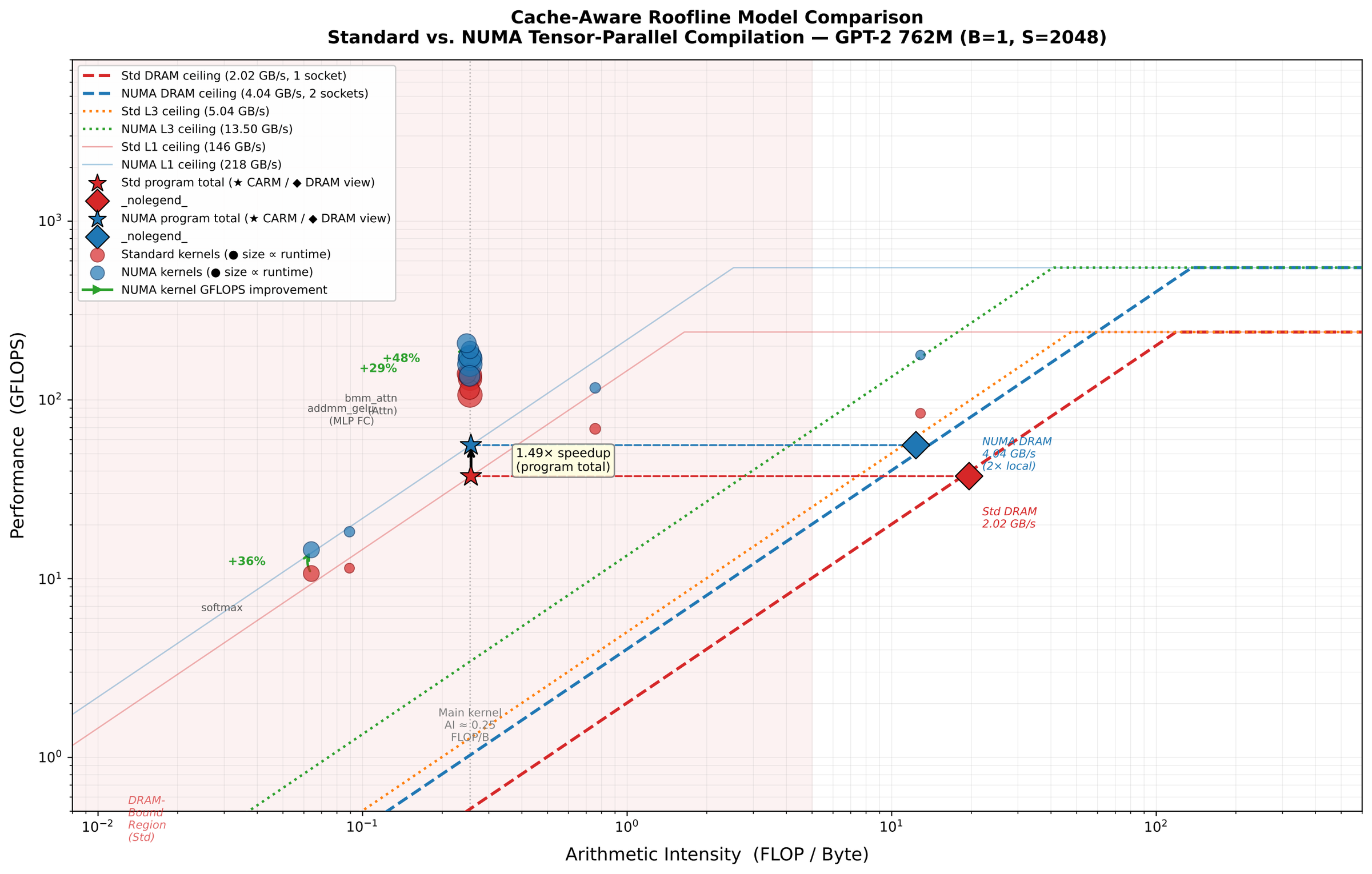

Cache-Aware Roofline Model Comparison
Standard vs. NUMA Tensor-Parallel Compilation — GPT-2 762M (B=1, S=2048)
Std DRAM ceiling (2.02 GB/s, 1 socket)
NUMA DRAM ceiling (4.04 GB/s, 2 sockets)
Std L3 ceiling (5.04 GB/s)
NUMA L3 ceiling (13.50 GB/s)
Std L1 ceiling (146 GB/s)
NUMA L1 ceiling (218 GB/s)
Std program total (★ CARM / ◆ DRAM view)
nolegend
NUMA program total (★ CARM / ◆ DRAM view)
nolegend
Standard kernels (● size ∝ runtime)
NUMA kernels (● size ∝ runtime)
NUMA kernel GFLOPS improvement
+48%
+29%
bmm_attn
addmm_gelu
(MLP FC)
(Attn)
+36%
softmax
1.49× speedup
(program total)
NUMA DRAM
4.04 GB/s
(2× local)
Std DRAM
2.02 GB/s
Main kernel
AI ≈ 0.25
FLOP/B
DRAM-
Bound
Region
(Std)
Performance (GFLOPS)
Arithmetic Intensity (FLOP / Byte)
10^3
10^2
10^1
10^0
10^-2
10^-1
10^0
10^1
10^2